

Process Management

Topic 3

Process Management

- Process Management is a major function of an operating system concerned with creating, scheduling, executing, synchronizing, and terminating processes.
- The operating system ensures that processes receive the necessary resources while maintaining efficient and safe execution.

Process Management

1. What is a Process?

A process is a program that is currently being executed.

A program is a passive set of instructions stored on a storage device, while a process is an active instance of that program.

Example:

When you open Microsoft Word:
Word program stored on disk



Loaded into RAM



Becomes a process



CPU executes it

A process contains:

- Program code
- Data
- Stack
- Heap
- CPU registers
- Program counter
- Process state

Process Management

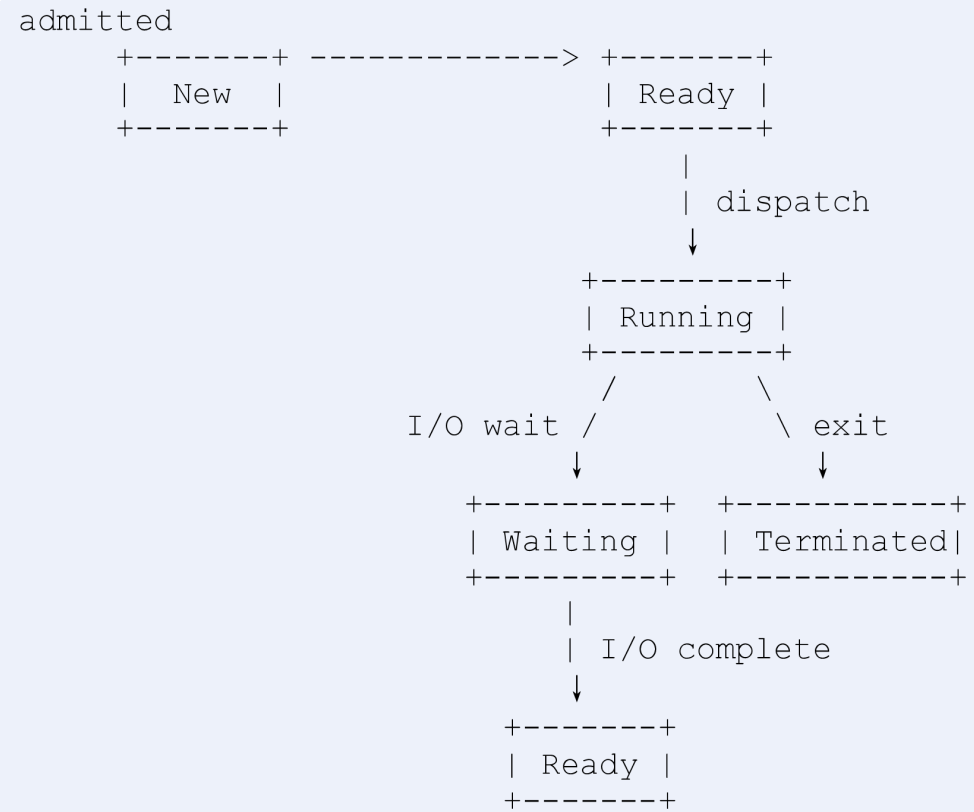
2. Process States

During its lifetime, a process normally moves through several states.

Main Process States

1. **New** – The process is being created.
2. **Ready** – The process is waiting to be assigned the CPU.
3. **Running** – Instructions are currently being executed by the CPU.
4. **Waiting/Blocked** – The process is waiting for an event, such as I/O completion.
5. **Terminated** – The process has finished execution.

Process-State Diagram



Process Management

3. Process Control Block (PCB)

The operating system maintains information about every process in a data structure called the Process Control Block (PCB).

A PCB typically contains:

Process ID (PID) – Unique identification number.

Process state – New, ready, running, waiting, or terminated.

Program counter – Address of the next instruction.

CPU registers – Current processor information.

CPU scheduling information – Priority and scheduling-related data.

Memory-management information – Information about memory allocated to the process.

Accounting information – CPU time used and other resource information.

I/O status information – Devices and files being used.

PCB Representation

```
+-----+
|           Process Control Block           |
+-----+
| Process ID                                |
| Process State                             |
| Program Counter                           |
| CPU Registers                             |
| Scheduling Information                     |
| Memory Information                         |
| Accounting Information                     |
| I/O Information                           |
+-----+
```

Process Management

4. Process Creation

A process can create another process.

The process that creates another process is called the parent process, while the newly created process is called the child process.

Parent Process

|

creates child

↓

Child Process

Process creation can occur when:

A user starts an application.

An existing process creates another process.

The operating system starts a system service.

A program requests creation of a new process.

5. Process Termination

A process terminates when it finishes its execution or when the operating system or another authorized process terminates it.

- Common reasons include:

Normal completion.

Error during execution.

User termination.

Operating-system termination.

Parent process termination.

- When a process terminates, the operating system releases the resources allocated to it.

Process Management

6. Process Scheduling

When several processes are ready to run, the operating system must determine which process should use the CPU next.

This is called CPU scheduling.

The scheduler selects a process from the ready queue and assigns it to the CPU.

Example

Ready Queue

P1 → P2 → P3 → P4

|



Scheduler

|



CPU

Important scheduling algorithms include:

- **First-Come, First-Served (FCFS)**
- **Shortest Job First (SJF)**
- **Shortest Remaining Time First (SRTF)**
- **Priority Scheduling**
- **Round Robin**
- **Multilevel Queue**
- **Multilevel Feedback Queue**

7. Context Switching

A context switch occurs when the CPU changes from executing **one process to executing another**.

- The operating system:
 1. Saves the state of the currently running process in its PCB.
 2. Selects another process.
 3. Loads the saved state of the new process.
 4. Resumes execution of the new process.

Process P1



Save P1 context



Load P2 context



Process P2

- Context switching allows multitasking, but it also introduces **overhead** because the CPU **spends time switching rather than executing** application instructions

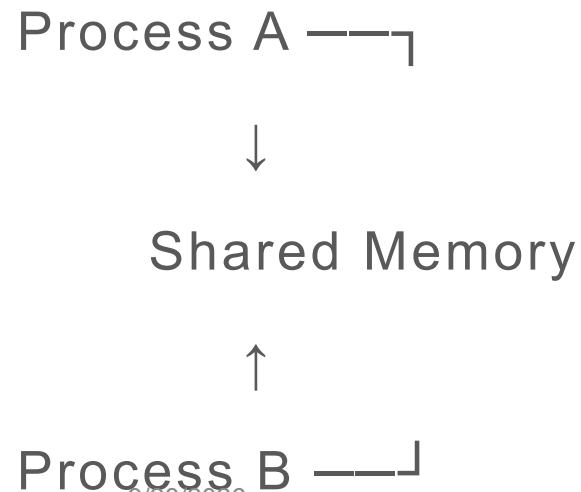
8. Inter-Process Communication (IPC)

Inter-Process Communication (IPC) allows processes to exchange information and coordinate their activities.

- **Two major IPC mechanisms are:**

- a. Shared Memory**

Processes communicate by accessing a common region of memory.



Inter-Process Communication (IPC)

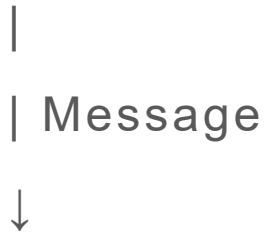
Advantage: Fast communication.

Disadvantage: Requires careful synchronization.

b. Message Passing

- Processes communicate by sending and receiving messages.

Process A



Process B

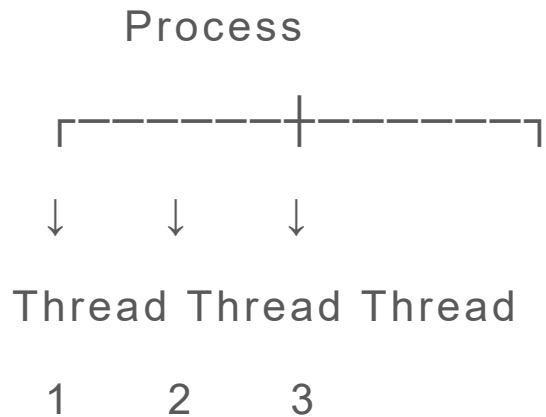
Advantage: Easier isolation between processes.

Disadvantage: Communication can be slower than shared memory.

9. Threads

A thread is the **smallest unit of CPU execution within a process**.

A process can contain one or multiple threads.



- **Advantages of Threads**

Improved responsiveness.

Resource sharing.

Lower overhead than creating separate processes.

Better utilization of multicore processors.

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Allows concurrent execution.

10. Process Synchronization

When multiple processes or threads access shared resources simultaneously, problems can occur.

Process synchronization ensures that processes coordinate their activities correctly.

- **Important concepts include:**

- Critical section**

- Race condition**

- Mutex**

- Semaphore**

- Monitor**

Example of a Race Condition

Suppose two processes simultaneously update the same bank account balance. If they access the data without proper synchronization, the final balance may be incorrect.

11. Process Queues

- The operating system maintains different queues for processes.

Job Queue

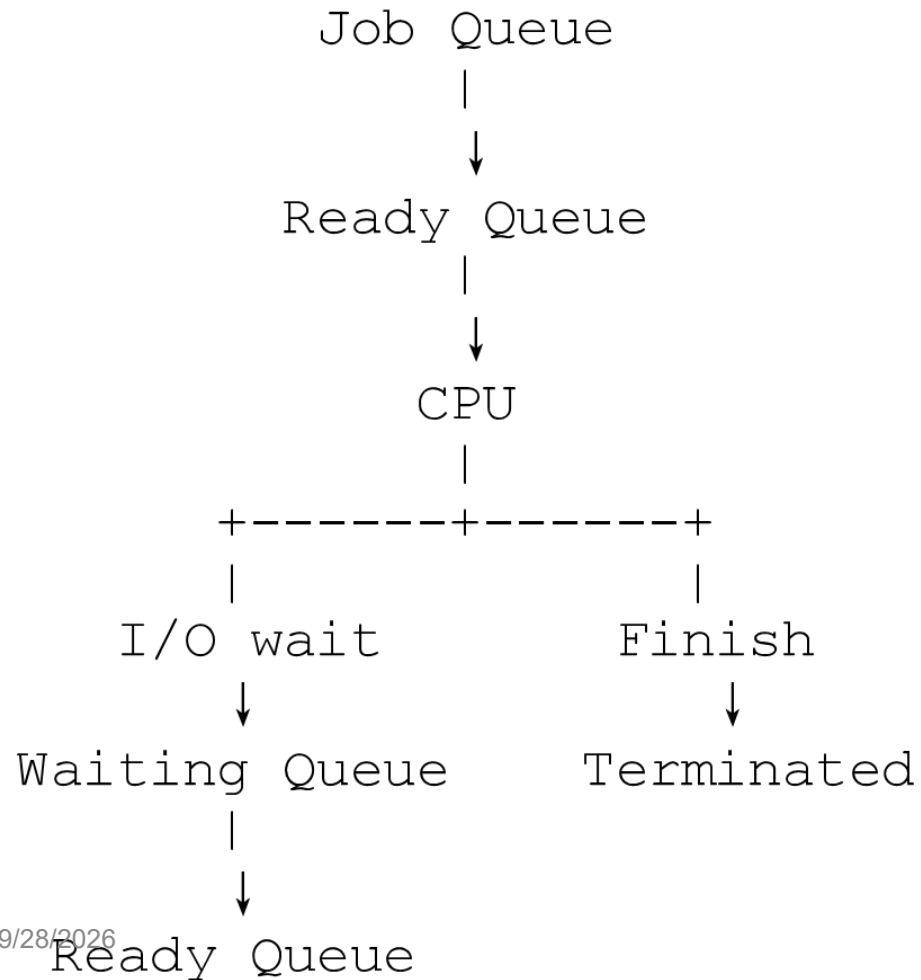
Contains all processes in the system.

● Ready Queue

Contains processes that are ready to execute.

Device/Waiting Queue

Contains processes waiting for a particular I/O device or event.



Process Management Responsibilities

The operating system is responsible for:

1. Creating and deleting processes.
2. Scheduling processes.
3. Suspending and resuming processes.
4. Managing process states.
5. Performing context switching.
6. Providing IPC mechanisms.
7. Providing process synchronization.
8. Allocating resources to processes.
9. Handling process termination.

Protecting processes from unauthorized access.

13. Process vs Program

Program

Passive

Stored on disk

Contains instructions

Does not have a process state

Can become a process

Process

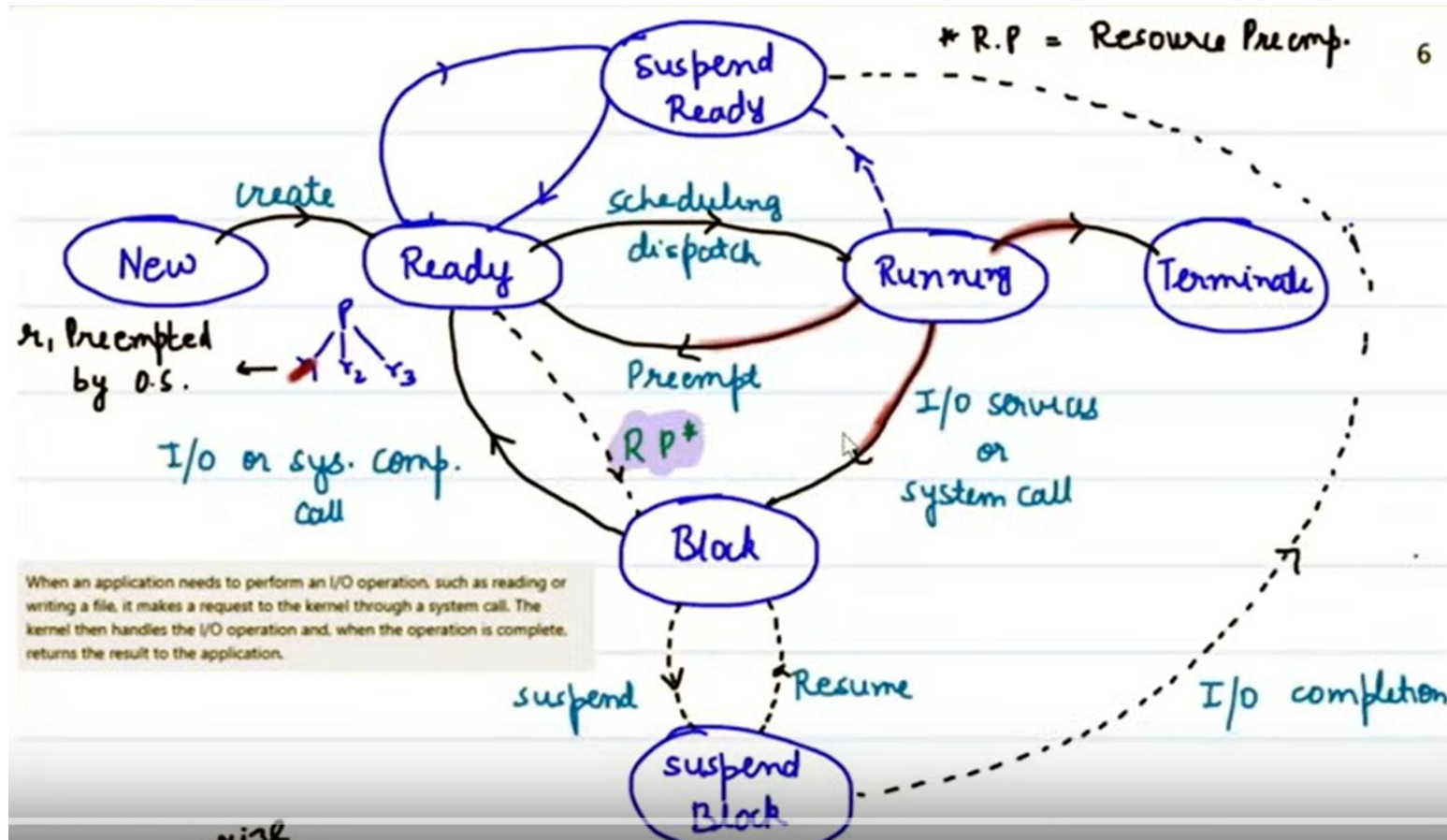
Active

Executing in memory

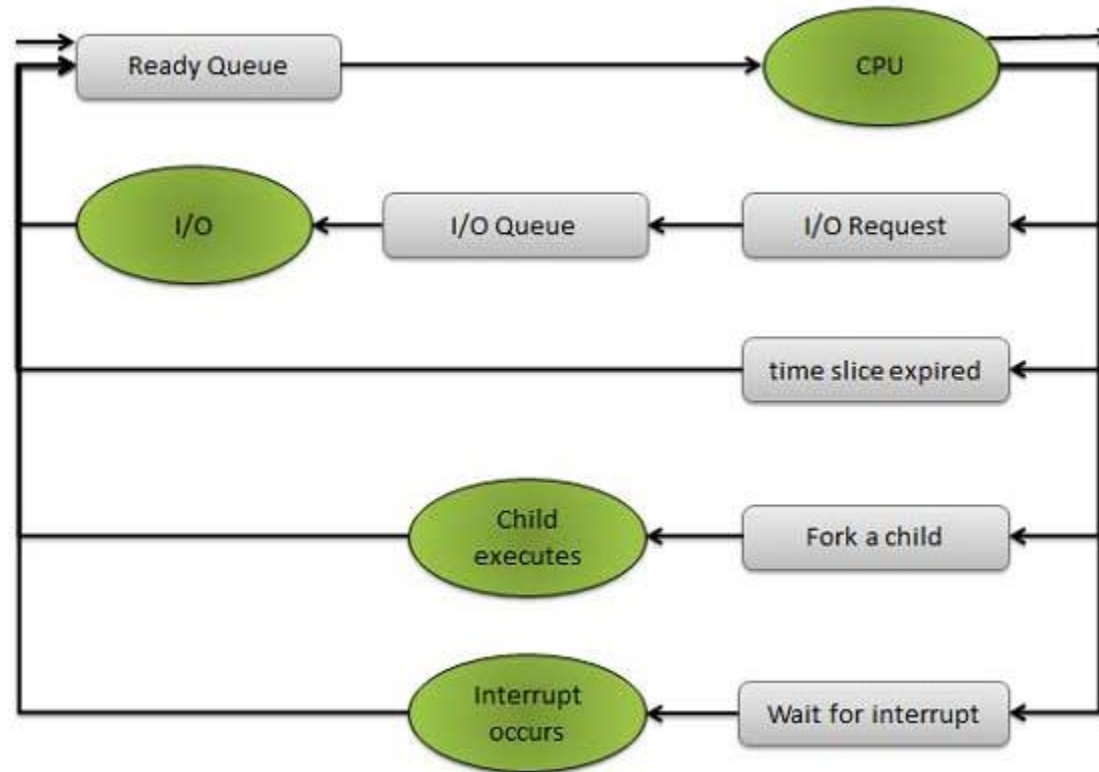
Contains instructions plus execution state

Has a process state

Is an executing instance of a program



Queuing diagram of process scheduling.



Process Scheduling

- A newly arrived process is put in the ready queue. Processes wait in ready queue for allocating the CPU. Once the CPU is assigned to a process, then that process will execute. While executing the process, any one of the following events can occur.
- The process could issue an I/O request and then it would be placed in an I/O queue.
- The process could create new sub process and will wait for its termination.
- The process could be removed forcibly from the CPU, as a result of interrupt and put back in the ready queue.

Long Term Scheduler/job scheduler

- Long Term Scheduler selects the processes from secondary storage and loads them into memory for execution.
- It is called “long term” because the time for which the scheduling is valid is long.
- The frequency of execution of a long term scheduler is usually low, as there may be minutes between the creation

Long Term Scheduler

- The primary objective of long term scheduler is to control the “degree of multiprogramming”.
- Degree of multiprogramming refers to the total number of processes present in the memory.
- If the degree of multiprogramming is stable, then the average rate of process creation is equal to the average terminate rate.

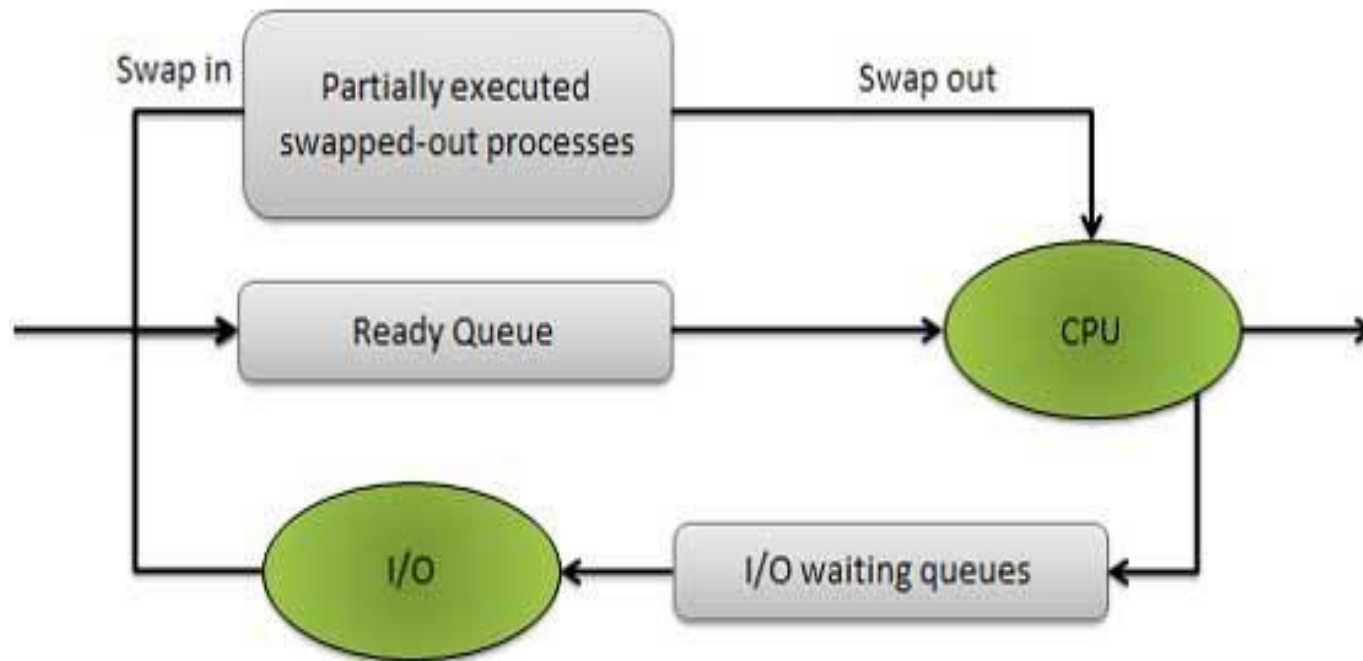
Long Term Scheduler

- This scheduler shows the best performance by selecting the good mixture of I/O bound and CPU bound processes.
- I/O bound processes are those that spend most of their time in I/O.
- CPU bound processes are those that spend most of their time in computations.

Medium Term Scheduler

- The medium term scheduler is required at the time when a swapped- out process is to be brought into pool of ready processes.
- A running process may be suspended because of I/O request.
- Such a suspended process is then removed from main memory and stored in secondary memory.

Medium Term Scheduler



Medium Term Scheduler

- This is done because there is a limit on the number of active processes that can reside in main memory.
- Therefore, a suspended process is swapped-out from main memory.
 - At some later time, the process can be swapped-in into the main memory.
 - All versions of Windows use swapping

Short Term Scheduler

- Short term scheduler selects one process from many ready processes that are residing in main memory and allocates CPU to one of them.
- Thus, it handles the scheduling of the processes that are in ready state.
- Short term scheduler is also known as
- CPU Scheduler.

Short Term Scheduler

- As compared to long term scheduler, a short term scheduler has to work very often.
- The frequency of execution of short term scheduler is high.
- It must select a new process for CPU

Comparison between Schedulers

S.N.	Long Term Scheduler	Short Term Scheduler	Medium Term Scheduler
1	It is a job scheduler	It is a CPU scheduler	It is a process swapping scheduler.
2	Speed is lesser than short term scheduler	Speed is fastest among other two	Speed is in between both short and long term scheduler.
3	It controls the degree of multiprogramming	It provides lesser control over degree of multiprogramming	It reduces the degree of multiprogramming.
4	It is almost absent or minimal in time sharing system	It is also minimal in time sharing system	It is a part of Time sharing systems.
5	It selects processes from pool and loads them into memory for execution	It selects those processes which are ready to execute	It can re-introduce the process into memory and execution can be continued.

Summary

Process management enables an operating system to manage multiple executing programs efficiently. It involves **process creation, process states, scheduling, context switching, IPC, synchronization, thread management, and process termination**. Effective process management is essential for achieving **multitasking, responsiveness, CPU utilization, and system stability**.

